

Quad11

A four-band bus compressor with an EQ and a saturator in every band.



Overview

Quad11 splits the signal into four bands - LOW, LO-MID, HI-MID and HIGH - and gives each its own bell EQ, compressor and saturator. The crossovers are steep (24 dB per octave) and the bands sum back flat, so with nothing switched on Quad11 is transparent. A master EQ sits on the whole bus ahead of the split.

Each band's compressor can also work the other way: turn THRESH above zero and it expands, lifting the peaks for extra punch. The level is made up automatically, so a band keeps its loudness while its peaks are held down or lifted. MIX blends the processed bus with the dry one for parallel compression, and the dry signal is kept exactly in time with the bands, so the blend is clean.

Reach for it for

- Mix bus and master glue, band by band
- Drum bus punch and parallel smash
- Taming harshness and esses on vocals without dulling the rest

- Tightening a bass without squashing its top
- Lifting transients (expansion) or adding tape-style warmth per band

Controls

Display

Response curve	The combined frequency response of the master EQ and the four band EQs, from 20 Hz to 20 kHz, -15 to +15 dB.
Crossover lines	The three splits between the bands. Drag a line to move it. The lowest can go down to 30 Hz, the highest up to 16 kHz, and the splits stay at least a quarter of their frequency apart.
Band dots	One dot per band in its colour: drag it to set the bell's frequency and gain; use the mouse wheel on it to change its width. Hovering shows the frequency, gain and Q.
Gain reduction	Each band's area is tinted, and its gain reduction (or expansion) hangs from the top.

MASTER EQ

Filters and shelves on the whole bus, ahead of the split.

HPF	A high-pass filter from OFF (fully left) up to 500 Hz.
LO FREQ / LO SHELF	A low shelf: its corner from 30 to 500 Hz and its gain, -12 to +12 dB.
LPF	A low-pass filter from 2 kHz up to OFF (fully right).
HI FREQ / HI SHELF	A high shelf: its corner from 2 to 16 kHz and its gain, -12 to +12 dB.

The four bands: LOW, LO-MID, HI-MID, HIGH

Each strip has the same controls, and shows the band's frequency range at the top right.

SOLO	Listen to this band on its own. Several bands can be soloed at once; while any band is soloed, MIX is ignored so you hear only the soloed bands, and SOLO ON shows in MASTER.
BYP	Lets the band through unprocessed (no EQ, compression or saturation).
Gain-reduction bar	Shows how much the band is being turned down (GR, up to 24 dB across the bar) or lifted (EXP, up to 6 dB).

EQ (per band)

FREQ	The bell's centre, 20 Hz to 20 kHz.
WIDTH	The bell's width (Q), 0.3 to 6. Higher is narrower.
GAIN	-12 to +12 dB.

COMPRESSOR (per band)

THRESH	From -48 to +48 dB; 0 is OFF. Turned left (COMP), it is the threshold in dBFS above which the band is compressed. Turned right (EXP), it lifts the peaks above minus that level (for example +18 lifts what goes over -18 dBFS), by up to 6 dB, for punch. Stereo-linked, with a 6 dB soft knee.
RATIO	How hard the compression or expansion works, 1:1 to 20:1.
ATTACK FAST / SLOW	How quickly the band reacts. The times suit each band: FAST is 10, 6, 3 and 1 ms from LOW to HIGH; SLOW is 40, 30, 20 and 10 ms.
RELEASE FAST / SLOW	How quickly the band recovers. FAST is 150, 120, 90 and 60 ms from LOW to HIGH; SLOW is 600, 450, 350 and 250 ms.

The make-up gain is automatic: it follows three quarters of the band's average gain change, slowly (over about 1.5 seconds), so the band keeps its loudness while the peaks are still held down or lifted.

SATURATION (per band)

TAPE / TUBE / CLIP	Click the key to step through the three types; its light is on while DRIVE is above zero. TAPE: a lopsided soft curve with even and odd harmonics. TUBE: one side gives way first - warmer, mostly even harmonics. CLIP: a hard knee, odd harmonics.
DRIVE	How hard the band is driven, 0 - 100% (up to +30 dB into the curve). It is level matched, so it changes the colour rather than the volume. While any band is driven the saturation runs at twice the sample rate to keep it clean.
PRE / POST	Where the saturation sits in the band (the light is on for PRE). PRE: saturation, then EQ, then the compressor - the compressor tames what the saturation adds. POST (default): EQ, compressor, then saturation - the compressor steadies the level going into the saturation.

INPUT

TRIM	Input level before anything else, from -24 to +12 dB. A single click on the knob (without dragging) switches GUARD on or off; its LED at the knob's upper right is dark when GUARD is off, red when it is on, and glows brighter while it is catching peaks.
GUARD	Off by default in this plugin. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.
IN meter	The input level after TRIM and GUARD, with a floating peak. Click a meter to clear the peak.

MASTER

MIX	Parallel compression: blends the processed bus with the dry one, 0 - 100%. 100% is fully processed.
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OUTPUT	Output level, -24 to +12 dB.
SAT	Band saturation on the way out, 0 - 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). The CLIP key flashes red while the clipper is working.
OUT meter	Output level, with a floating peak.

The header

PRESETS, arrows and display	PRESETS (or a click on the display) opens the list of factory presets; the ◀ / ▶ keys step through them. A preset sets the crossovers, the master EQ, all four bands and MIX; TRIM, GUARD, OUTPUT, SAT and the OUT stage are left as they are. The last entry resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Gentle Glue	All four bands at -14 dB, 1.5:1, slow attack and release: light, even bus glue.
Mix Bus Punch	HPF at 25 Hz; 2:1 to 3:1 on every band with slow attack and fast release, and a little tape drive on the low band.
Drum Bus Smash (parallel)	Heavy, fast compression (-30 dB, 6:1) on every band, tape and clip saturation, a touch of low and high shelf, blended in at 45%.

Transient Lift (expand)	All bands expanding (THRESH above zero) with fast attack and release, to lift the hits.
Vocal Control	Bands split at 200 Hz, 2 kHz and 7 kHz, HPF at 80 Hz; firmer, faster compression on the upper bands to tame harshness and esses.
Bass Tighten	Bands split at 80 Hz, 250 Hz and 2 kHz; the two low bands compressed hard with tube drive before the EQ, a little low-mid cut, the top band left alone.
De-Mud & Air	A 3 dB cut at 350 Hz in the low-mid band, a lift at 12 kHz and an air shelf, with gentle compression throughout.
Tape Warmth	No compression: tape saturation on every band, more on the lows than the highs.
Master Gentle	Very light, slow compression (1.3:1 to 1.5:1) with a little tape drive on the low band.
Parallel Crush	Every band crushed (-36 dB, 10:1, fast) and clipped, blended in at 35%.
Reset all to defaults	Every control back to its starting value.

Tips

- Solo a band while you set its crossover lines, so you hear exactly what it holds.
- Start with THRESH just below where the band's GR bar starts moving, then raise RATIO until it sits.
- For a parallel bus, crush the bands hard and bring MIX down to 30 - 50% - the dry signal stays perfectly in time with them.
- Use PRE saturation when you want the compressor to smooth out the drive, POST when you want the drive to sit on a steady level.
- Expansion (THRESH above zero) on the low and hi-mid bands brings back the snap of over-compressed drums.

File: gfx-quad11.jsfx